

Лабораторная работа №5

Дисциплина: Сетевые технологии

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2026-03-20

1 Цель работы

- Построение простейших моделей сетей в GNS3

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- Построение простейших моделей сетей в GNS3
- Использование коммутатора и маршрутизаторов FRR и VyOS

1 Цель работы

- Построение простейших моделей сетей в GNS3
- Использование коммутатора и маршрутизаторов FRR и VyOS
- Анализ сетевого трафика с помощью Wireshark

2 Запуск GNS3 и создание топологии

- Запущены GNS3 VM и GNS3

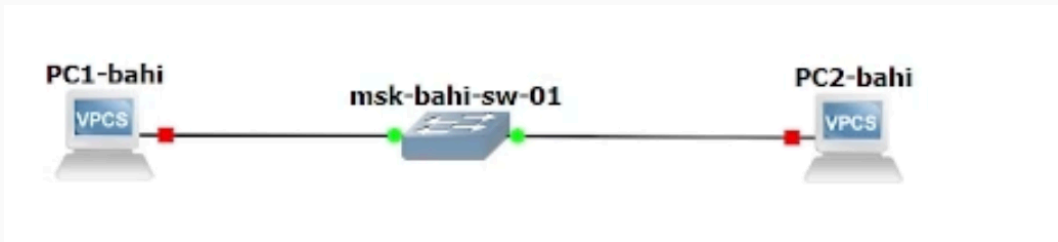


Рисунок 1: Топология простейшей сети в GNS3

2 Запуск GNS3 и создание топологии

- Запущены GNS3 VM и GNS3
- Создан новый проект

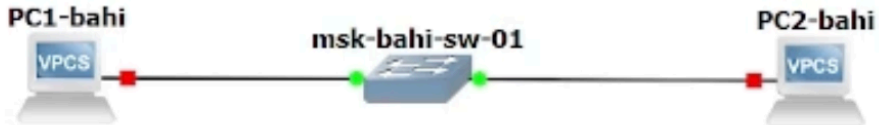


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2 Запуск GNS3 и создание топологии

- Запущены GNS3 VM и GNS3
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- Запущены GNS3 VM и GNS3
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- Устройства переименованы по шаблону



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2 Запуск GNS3 и создание топологии

- Запущены GNS3 VM и GNS3
- Создан новый проект
- Размещены коммутатор Ethernet и два узла VPCS
- Устройства переименованы по шаблону
- Отображены интерфейсы соединений

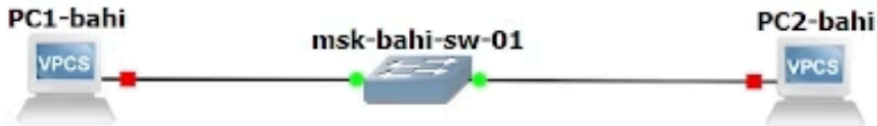


Рисунок 1: Топология простейшей сети в GNS3

3 Настройка IP-адресации PC-1

- Открыт терминал PC-1

```
Press '?' to get help.
```

```
Executing the startup file
```

```
PC1-bahi>
```

```
PC1-bahi>
```

```
PC1-bahi>
```

```
PC1-bahi> /?
```

```
?
```

```
Print help
```

```
! COMMAND [ARG ...]
```

```
Invoke an OS COMMAND with optional ARG(s)
```

```
arp
```

```
Shortcut for: show arp. Show arp table
```

```
clear ARG
```

```
Clear IPv4/IPv6, arp/neighbor cache, command history
```

```
dhcp [OPTION]
```

```
Shortcut for: ip dhcp. Get IPv4 address via DHCP
```

```
disconnect
```

```
Exit the telnet session (daemon mode)
```

```
echo TEXT
```

```
Display TEXT in output. See also set echo ?
```

```
help
```

```
Print help
```

```
history
```

```
Shortcut for: show history. List the command history
```

```
ip ARG ... [OPTION]
```

```
Configure the current VPC's IP settings. See ip ?
```

```
load [FILENAME]
```

```
Load the configuration/script from the file FILENAME
```

```
ping HOST [OPTION ...]
```

```
Ping HOST with ICMP (default) or TCP/UDP. See ping ?
```

```
quit
```

```
Quit program
```

```
relay ARG ...
```

```
Configure packet relay between UDP ports. See relay ?
```

3 Настройка IP-адресации PC-1

- Открыт терминал PC-1
- Назначен IPv4-адрес 192.168.1.11/24

```
Press '?' to get help.
```

```
Executing the startup file
```

```
PC1-bahi>
```

```
PC1-bahi>
```

```
PC1-bahi>
```

```
PC1-bahi> /?
```

```
?
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```
Quit program
```

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Configure packet relay between UDP ports. See relay ?
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3 Настройка IP-адресации PC-1

- Открыт терминал PC-1
- Назначен IPv4-адрес 192.168.1.11/24
- Указан шлюз 192.168.1.1

```
Press '?' to get help.
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```
Executing the startup file
```

```
PC1-bahi>
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```
PC1-bahi>
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```
PC1-bahi>
```

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PC1-bahi> /?
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! COMMAND [ARG ...]
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clear ARG
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3 Настройка IP-адресации PC-1

- Открыт терминал PC-1
- Назначен IPv4-адрес 192.168.1.11/24
- Указан шлюз 192.168.1.1
- Конфигурация сохранена

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Press '?' to get help.
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Executing the startup file
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```
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```

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PC1-bahi>
```

```
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Ping HOST with ICMP (default) or TCP/UDP. See ping ?
```

```
Quit program
```

```
Configure packet relay between UDP ports. See relay ?
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4 Настройка IP-адресации PC-2

- Назначен IPv4-адрес 192.168.1.12/24

```
ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

PC2-bahi> ping 192.168.1.12
192.168.1.12 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=5 ttl=64 time=0.001 ms
```

4 Настройка IP-адресации PC-2

- Назначен IPv4-адрес 192.168.1.12/24
- Выполнена проверка доступности PC-1

```
ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

PC2-bahi> ping 192.168.1.12
192.168.1.12 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=5 ttl=64 time=0.001 ms
```

4 Настройка IP-адресации PC-2

- Назначен IPv4-адрес 192.168.1.12/24
- Выполнена проверка доступности PC-1
- Получен ICMP эхо-ответ

```
ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

PC2-bahi> ping 192.168.1.12
192.168.1.12 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=5 ttl=64 time=0.001 ms
```

5 Проверка связи между PC-1 и PC-2

- Выполнен ping с PC-1 на PC-2

```
ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

PC2-bahi> ping 192.168.1.12
192.168.1.12 icmp_seq=1 ttl=64 time=0.001 ms
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192.168.1.12 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=5 ttl=64 time=0.001 ms
```

5 Проверка связи между PC-1 и PC-2

- Выполнен ping с PC-1 на PC-2
- Получены 5 ICMP-ответов

```
ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

PC2-bahi> ping 192.168.1.12
192.168.1.12 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=5 ttl=64 time=0.001 ms
```

5 Проверка связи между PC-1 и PC-2

- Выполнен ping с PC-1 на PC-2
- Получены 5 ICMP-ответов
- Связность подтверждена

```
ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

PC2-bahi> ping 192.168.1.12
192.168.1.12 icmp_seq=1 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=2 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=3 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=4 ttl=64 time=0.001 ms
192.168.1.12 icmp_seq=5 ttl=64 time=0.001 ms
```

6 Остановка узлов проекта

- Все устройства проекта остановлены

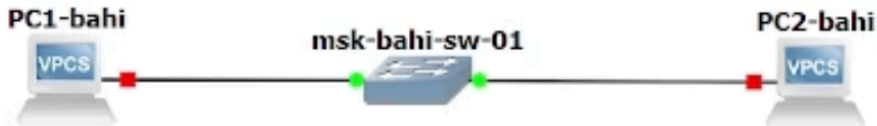


Рисунок 5: Остановка всех узлов

6 Остановка узлов проекта

- Все устройства проекта остановлены
- Подготовка к захвату трафика

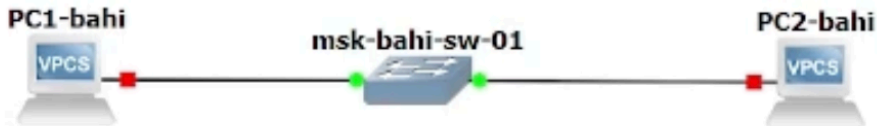


Рисунок 5: Остановка всех узлов

7 Запуск захвата трафика в Wireshark

- Включён Start capture на линии PC-1 ↔ коммутатор

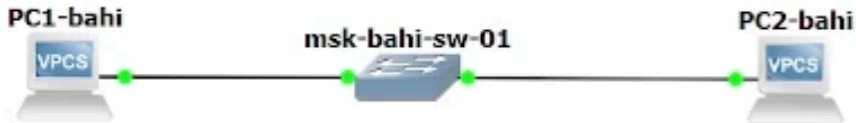


Рисунок 6: Захват трафика, старт узлов

7 Запуск захвата трафика в Wireshark

- Включён Start capture на линии PC-1 ↔ коммутатор
- Запущены все узлы

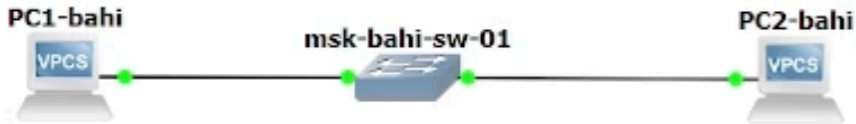


Рисунок 6: Захват трафика, старт узлов

7 Запуск захвата трафика в Wireshark

- Включён Start capture на линии PC-1 ↔ коммутатор
- Запущены все узлы
- Захват трафика активен

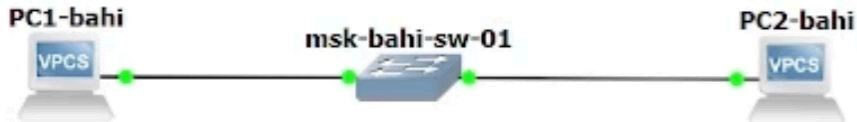


Рисунок 6: Захват трафика, старт узлов

8 Анализ ARP-пакетов

- Зафиксированы ARP / Gratuitous ARP

Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
4	1.051158	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1.104.848420	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
6	1.104.849418	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
→ 7	1.104.862896	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaca2, seq=1/256
← 8	1.104.863827	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaca2, seq=1/256
9	1.105.894632	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xada2, seq=2/512
10	1.105.895564	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xada2, seq=2/512
11	1.106.912002	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaea2, seq=3/768
12	1.106.912984	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaea2, seq=3/768
13	1.107.931469	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xafa2, seq=4/1024
14	1.107.932469	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xafa2, seq=4/1024
15	1.108.951009	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xb0a2, seq=5/1280
[Time delta from previous captured frame: 13.478000 milliseconds]						
[Time delta from previous displayed frame: 13.478000 milliseconds]						
[Time since reference or first frame: 18 minutes, 24.862896000 seconds]						
Frame Number: 7						
Frame Length: 98 bytes (784 bits)						
Capture Length: 98 bytes (784 bits)						
				0000	00 50 79 66 68 00 00 50 79 66 00 00 00 00 00 00	
				0010	00 54 a2 ac 00 00 40 01 54 95 00 00 00 00 00 00	
				0020	01 0b 08 00 73 68 ac a2 00 01 00 00 00 00 00 00	
				0030	0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d	
				0040	1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d	
				0050	2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d	
				0060	3e 3f	

8 Анализ ARP-пакетов

- Зафиксированы ARP / Gratuitous ARP
- MAC назначения: broadcast ff:ff:ff:ff:ff:ff

Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
4	1.051158	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1.048420	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
6	1.049418	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
→ 7	1.04862896	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaca2, seq=1/256
← 8	1.04863827	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaca2, seq=1/256
9	1.05894632	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xada2, seq=2/512
10	1.05895564	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xada2, seq=2/512
11	1.06912002	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaea2, seq=3/768
12	1.06912984	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaea2, seq=3/768
13	1.07931469	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xafa2, seq=4/1024
14	1.07932469	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xafa2, seq=4/1024
15	1.08951009	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xb0a2, seq=5/1280
[Time delta from previous captured frame: 13.478000 milliseconds]						
[Time delta from previous displayed frame: 13.478000 milliseconds]						
[Time since reference or first frame: 18 minutes, 24.862896000 seconds]						
Frame Number: 7						
Frame Length: 98 bytes (784 bits)						
Capture Length: 98 bytes (784 bits)						
				0000	00 50 79 66 68 00 00 50 79 66 00 00 00 00 00 00	
				0010	00 54 a2 ac 00 00 40 01 54 95 00 00 00 00 00 00	
				0020	01 0b 08 00 73 68 ac a2 00 01 00 00 00 00 00 00	
				0030	0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d	
				0040	1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d	
				0050	2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d	
				0060	3e 3f	

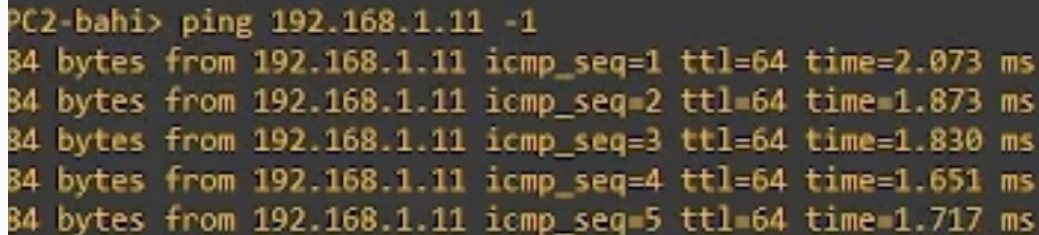
8 Анализ ARP-пакетов

- Зафиксированы ARP / Gratuitous ARP
- MAC назначения: broadcast ff:ff:ff:ff:ff:ff
- Длина кадра: 64 байта

Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
4	1.051158	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
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12	1.06912984	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaea2, seq=3/768
13	1.07931469	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xafa2, seq=4/1024
14	1.07932469	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xafa2, seq=4/1024
15	1.08951009	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xb0a2, seq=5/1280
[Time delta from previous captured frame: 13.478000 milliseconds]						
[Time delta from previous displayed frame: 13.478000 milliseconds]						
[Time since reference or first frame: 18 minutes, 24.862896000 seconds]						
Frame Number: 7						
Frame Length: 98 bytes (784 bits)						
Capture Length: 98 bytes (784 bits)						
				0000	00 50 79 66 68 00 00 50 79 66 00 00 00 00 00 00	
				0010	00 54 a2 ac 00 00 40 01 54 95 00 00 00 00 00 00	
				0020	01 0b 08 00 73 68 ac a2 00 01 00 00 00 00 00 00	
				0030	0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d	
				0040	1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d	
				0050	2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d	
				0060	3e 3f	

9 ICMP Echo Request (PC-2 → PC-1)

- Выполнен одиночный ICMP-запрос

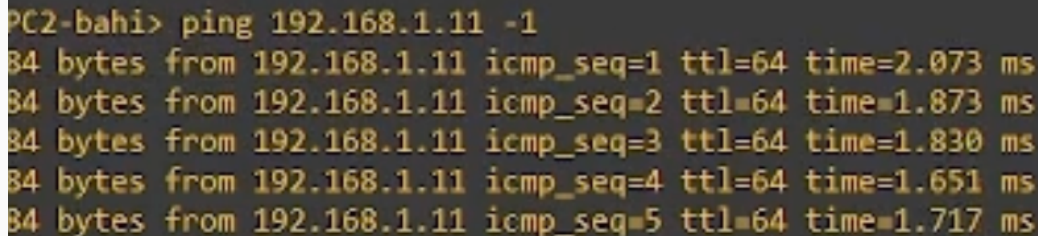


```
PC2-bahi> ping 192.168.1.11 -1
34 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=2.073 ms
34 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=1.873 ms
34 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.830 ms
34 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.651 ms
34 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.717 ms
```

Рисунок 8: Эхо-запрос в ICMP-модe

9 ICMP Echo Request (PC-2 → PC-1)

- Выполнен одиночный ICMP-запрос
- Использована опция -1

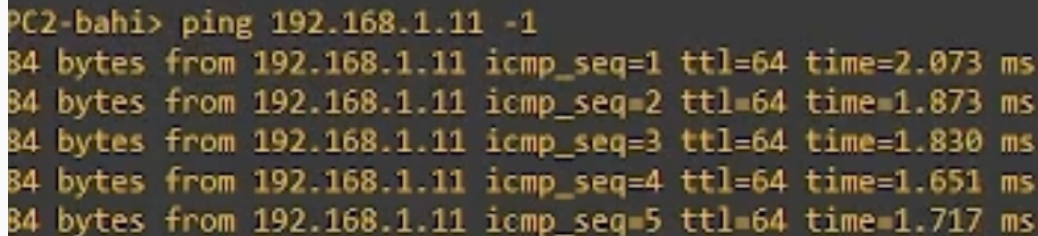


```
PC2-bahi> ping 192.168.1.11 -1
34 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=2.073 ms
34 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=1.873 ms
34 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.830 ms
34 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.651 ms
34 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.717 ms
```

Рисунок 8: Эхо-запрос в ICMP-моде

9 ICMP Echo Request (PC-2 → PC-1)

- Выполнен одиночный ICMP-запрос
- Использована опция -1
- Проверка базовой связности



```
PC2-bahi> ping 192.168.1.11 -1
34 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=2.073 ms
34 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=1.873 ms
34 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.830 ms
34 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=1.651 ms
34 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.717 ms
```

Рисунок 8: Эхо-запрос в ICMP-модe

10 Анализ ICMP в Wireshark

- MAC-адреса: unicast

Capturing from Standard input [PC1-bah1 Ethernet0 to msk-bah1-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
4	1.051158	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1104.848420	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
6	1104.849418	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
7	1104.862896	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaca2, seq=1/256, ttl=64
8	1104.863827	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaca2, seq=1/256, ttl=64
9	1105.894632	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xada2, seq=2/512, ttl=64
10	1105.895564	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xada2, seq=2/512, ttl=64
11	1106.912002	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaea2, seq=3/768, ttl=64
12	1106.912984	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaea2, seq=3/768, ttl=64
13	1107.931469	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xafa2, seq=4/1024, ttl=64
14	1107.932469	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xafa2, seq=4/1024, ttl=64
15	1108.951009	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xb0a2, seq=5/1280, ttl=64

Destination Address: 192.168.1.11
[Stream index: 0]

Internet Control Message Protocol
Type: Echo (ping) request (8)

0000 00 50 79 66 68 00 00 50 79 66 68 01 00
0010 00 54 a2 ac 00 00 40 01 54 95 c0 a8 01
0020 01 0b 08 00 73 68 ac a2 00 01 08 09 0e
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a
0040 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27 28 29

10 Анализ ICMP в Wireshark

- MAC-адреса: unicast
- Протокол: ICMP

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
4	1.051158	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1104.848420	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
6	1104.849418	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
→ 7	1104.862896	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaca2, seq=1/256, ttl=64
← 8	1104.863827	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaca2, seq=1/256, ttl=64
9	1105.894632	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xada2, seq=2/512, ttl=64
10	1105.895564	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xada2, seq=2/512, ttl=64
11	1106.912002	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaea2, seq=3/768, ttl=64
12	1106.912984	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaea2, seq=3/768, ttl=64
13	1107.931469	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xafa2, seq=4/1024, ttl=64
14	1107.932469	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xafa2, seq=4/1024, ttl=64
15	1108.951009	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xb0a2, seq=5/1280, ttl=64

Destination Address: 192.168.1.11
[Stream index: 0]

Internet Control Message Protocol
Type: Echo (ping) request (8)

0000 00 50 79 66 68 00 00 50 79 66 68 01 00
0010 00 54 a2 ac 00 00 40 01 54 95 c0 a8 01
0020 01 0b 08 00 73 68 ac a2 00 01 08 09 0e
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a
0040 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27 28 29

10 Анализ ICMP в Wireshark

- MAC-адреса: unicast
- Протокол: ICMP
- Источник: 192.168.1.12

Capturing from Standard input [PC1-bah1 Ethernet0 to msk-bah1-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
4	1.051158	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1104.848420	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
6	1104.849418	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
7	1104.862896	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaca2, seq=1/256, ttl=64
8	1104.863827	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaca2, seq=1/256, ttl=64
9	1105.894632	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xada2, seq=2/512, ttl=64
10	1105.895564	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xada2, seq=2/512, ttl=64
11	1106.912002	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaea2, seq=3/768, ttl=64
12	1106.912984	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaea2, seq=3/768, ttl=64
13	1107.931469	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xafa2, seq=4/1024, ttl=64
14	1107.932469	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xafa2, seq=4/1024, ttl=64
15	1108.951009	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xb0a2, seq=5/1280, ttl=64

Destination Address: 192.168.1.11
[Stream index: 0]

Internet Control Message Protocol
Type: Echo (ping) request (8)

0000 00 50 79 66 68 00 00 50 79 66 68 01 00
0010 00 54 a2 ac 00 00 40 01 54 95 c0 a8 01
0020 01 0b 08 00 73 68 ac a2 00 01 08 09 0e
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a
0040 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27 28 29

10 Анализ ICMP в Wireshark

- MAC-адреса: unicast
- Протокол: ICMP
- Источник: 192.168.1.12
- Назначение: 192.168.1.11

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
4	1.051158	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.11 (Request)
5	1.104.848420	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.11? Tell 192.168.1.12
6	1.104.849418	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.11 is at 00:50:79:66:68:00
7	1.104.862896	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaca2, seq=1/256, ttl=64
8	1.104.863827	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaca2, seq=1/256, ttl=64
9	1.105.894632	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xada2, seq=2/512, ttl=64
10	1.105.895564	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xada2, seq=2/512, ttl=64
11	1.106.912002	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xaea2, seq=3/768, ttl=64
12	1.106.912984	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xaea2, seq=3/768, ttl=64
13	1.107.931469	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xafa2, seq=4/1024, ttl=64
14	1.107.932469	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0xafa2, seq=4/1024, ttl=64
15	1.108.951009	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0xb0a2, seq=5/1280, ttl=64

Destination Address: 192.168.1.11
[Stream index: 0]

Internet Control Message Protocol
Type: Echo (ping) request (8)

0000 00 50 79 66 68 00 00 50 79 66 68 01 00
0010 00 54 a2 ac 00 00 40 01 54 95 c0 a8 01
0020 01 0b 08 00 73 68 ac a2 00 01 08 09 0e
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a
0040 1c 1d 1e 1f 20 21 22 23 24 25 26 27 28 29

11 UDP Echo Request

- Выполнен UDP-эхо-запрос

```
PC2-bahi>  
PC2-bahi> ping 192.168.1.11 -2  
84 bytes from 192.168.1.11 udp_seq=1 ttl=64 time=1.841 ms  
84 bytes from 192.168.1.11 udp_seq=2 ttl=64 time=1.552 ms  
84 bytes from 192.168.1.11 udp_seq=3 ttl=64 time=1.661 ms  
84 bytes from 192.168.1.11 udp_seq=4 ttl=64 time=1.922 ms  
84 bytes from 192.168.1.11 udp_seq=5 ttl=64 time=1.754 ms  
PC2-bahi> █
```

Рисунок 9: Эхо-запрос в UDP-моде

11 UDP Echo Request

- Выполнен UDP-эхо-запрос
- Использована опция -2

```
PC2-bahi>  
PC2-bahi> ping 192.168.1.11 -2  
84 bytes from 192.168.1.11 udp_seq=1 ttl=64 time=1.841 ms  
84 bytes from 192.168.1.11 udp_seq=2 ttl=64 time=1.552 ms  
84 bytes from 192.168.1.11 udp_seq=3 ttl=64 time=1.661 ms  
84 bytes from 192.168.1.11 udp_seq=4 ttl=64 time=1.922 ms  
84 bytes from 192.168.1.11 udp_seq=5 ttl=64 time=1.754 ms  
PC2-bahi>
```

Рисунок 9: Эхо-запрос в UDP-моде

11 UDP Echo Request

- Выполнен UDP-эхо-запрос
- Использована опция -2
- Проверка работы UDP

```
PC2-bahi>  
PC2-bahi> ping 192.168.1.11 -2  
84 bytes from 192.168.1.11 udp_seq=1 ttl=64 time=1.841 ms  
84 bytes from 192.168.1.11 udp_seq=2 ttl=64 time=1.552 ms  
84 bytes from 192.168.1.11 udp_seq=3 ttl=64 time=1.661 ms  
84 bytes from 192.168.1.11 udp_seq=4 ttl=64 time=1.922 ms  
84 bytes from 192.168.1.11 udp_seq=5 ttl=64 time=1.754 ms  
PC2-bahi> █
```

Рисунок 9: Эхо-запрос в UDP-моде

12 Анализ UDP-пакетов

- Протокол: UDP

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
27	1493.149960	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x30a4, seq=5/1280,
28	1493.150930	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x30a4, seq=5/1280,
29	1528.534198	192.168.1.12	192.168.1.11	ECHO	98	Request
30	1528.535130	192.168.1.11	192.168.1.12	ECHO	98	Response
31	1529.564271	192.168.1.12	192.168.1.11	ECHO	98	Request
32	1529.565276	192.168.1.11	192.168.1.12	ECHO	98	Response
33	1530.580318	192.168.1.12	192.168.1.11	ECHO	98	Request
34	1530.581300	192.168.1.11	192.168.1.12	ECHO	98	Response
35	1531.598809	192.168.1.12	192.168.1.11	ECHO	98	Request
36	1531.600296	192.168.1.11	192.168.1.12	ECHO	98	Response
37	1532.619838	192.168.1.12	192.168.1.11	ECHO	98	Request
38	1532.620862	192.168.1.11	192.168.1.12	ECHO	98	Response

12 Анализ UDP-пакетов

- Протокол: UDP
- Порт источника: динамический

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
27	1493.149960	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x30a4, seq=5/1280,
28	1493.150930	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x30a4, seq=5/1280,
29	1528.534198	192.168.1.12	192.168.1.11	ECHO	98	Request
30	1528.535130	192.168.1.11	192.168.1.12	ECHO	98	Response
31	1529.564271	192.168.1.12	192.168.1.11	ECHO	98	Request
32	1529.565276	192.168.1.11	192.168.1.12	ECHO	98	Response
33	1530.580318	192.168.1.12	192.168.1.11	ECHO	98	Request
34	1530.581300	192.168.1.11	192.168.1.12	ECHO	98	Response
35	1531.598809	192.168.1.12	192.168.1.11	ECHO	98	Request
36	1531.600296	192.168.1.11	192.168.1.12	ECHO	98	Response
37	1532.619838	192.168.1.12	192.168.1.11	ECHO	98	Request
38	1532.620862	192.168.1.11	192.168.1.12	ECHO	98	Response

12 Анализ UDP-пакетов

- Протокол: UDP
- Порт источника: динамический
- Порт назначения: 7 (Echo)

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
27	1493.149960	192.168.1.12	192.168.1.11	ICMP	98	Echo (ping) request id=0x30a4, seq=5/1280,
28	1493.150930	192.168.1.11	192.168.1.12	ICMP	98	Echo (ping) reply id=0x30a4, seq=5/1280,
29	1528.534198	192.168.1.12	192.168.1.11	ECHO	98	Request
30	1528.535130	192.168.1.11	192.168.1.12	ECHO	98	Response
31	1529.564271	192.168.1.12	192.168.1.11	ECHO	98	Request
32	1529.565276	192.168.1.11	192.168.1.12	ECHO	98	Response
33	1530.580318	192.168.1.12	192.168.1.11	ECHO	98	Request
34	1530.581300	192.168.1.11	192.168.1.12	ECHO	98	Response
35	1531.598809	192.168.1.12	192.168.1.11	ECHO	98	Request
36	1531.600296	192.168.1.11	192.168.1.12	ECHO	98	Response
37	1532.619838	192.168.1.12	192.168.1.11	ECHO	98	Request
38	1532.620862	192.168.1.11	192.168.1.12	ECHO	98	Response

13 TCP Echo Request

- Выполнен TCP-эхо-запрос

```
PC2-bahi> ping 192.168.1.11 -3
Connect    7@192.168.1.11 seq=1 ttl=64 time=15.011 ms
SendData   7@192.168.1.11 seq=1 ttl=64 time=14.730 ms
Close      7@192.168.1.11 seq=1 ttl=64 time=30.098 ms
Connect    7@192.168.1.11 seq=2 ttl=64 time=14.951 ms
SendData   7@192.168.1.11 seq=2 ttl=64 time=15.139 ms
Close      7@192.168.1.11 seq=2 ttl=64 time=30.007 ms
Connect    7@192.168.1.11 seq=3 ttl=64 time=15.310 ms
SendData   7@192.168.1.11 seq=3 ttl=64 time=15.319 ms
Close      7@192.168.1.11 seq=3 ttl=64 time=30.571 ms
Connect    7@192.168.1.11 seq=4 ttl=64 time=15.066 ms
SendData   7@192.168.1.11 seq=4 ttl=64 time=15.542 ms
Close      7@192.168.1.11 seq=4 ttl=64 time=30.630 ms
```

13 TCP Echo Request

- Выполнен TCP-эхо-запрос
- Использована опция -3

```
PC2-bahi> ping 192.168.1.11 -3
Connect    7@192.168.1.11 seq=1 ttl=64 time=15.011 ms
SendData   7@192.168.1.11 seq=1 ttl=64 time=14.730 ms
Close      7@192.168.1.11 seq=1 ttl=64 time=30.098 ms
Connect    7@192.168.1.11 seq=2 ttl=64 time=14.951 ms
SendData   7@192.168.1.11 seq=2 ttl=64 time=15.139 ms
Close      7@192.168.1.11 seq=2 ttl=64 time=30.007 ms
Connect    7@192.168.1.11 seq=3 ttl=64 time=15.310 ms
SendData   7@192.168.1.11 seq=3 ttl=64 time=15.319 ms
Close      7@192.168.1.11 seq=3 ttl=64 time=30.571 ms
Connect    7@192.168.1.11 seq=4 ttl=64 time=15.066 ms
SendData   7@192.168.1.11 seq=4 ttl=64 time=15.542 ms
Close      7@192.168.1.11 seq=4 ttl=64 time=30.630 ms
```

13 TCP Echo Request

- Выполнен TCP-эхо-запрос
- Использована опция -3
- Установлено TCP-соединение

```
PC2-bah1> ping 192.168.1.11 -3
Connect    7@192.168.1.11 seq=1 ttl=64 time=15.011 ms
SendData   7@192.168.1.11 seq=1 ttl=64 time=14.730 ms
Close      7@192.168.1.11 seq=1 ttl=64 time=30.098 ms
Connect    7@192.168.1.11 seq=2 ttl=64 time=14.951 ms
SendData   7@192.168.1.11 seq=2 ttl=64 time=15.139 ms
Close      7@192.168.1.11 seq=2 ttl=64 time=30.007 ms
Connect    7@192.168.1.11 seq=3 ttl=64 time=15.310 ms
SendData   7@192.168.1.11 seq=3 ttl=64 time=15.319 ms
Close      7@192.168.1.11 seq=3 ttl=64 time=30.571 ms
Connect    7@192.168.1.11 seq=4 ttl=64 time=15.066 ms
SendData   7@192.168.1.11 seq=4 ttl=64 time=15.542 ms
Close      7@192.168.1.11 seq=4 ttl=64 time=30.630 ms
```

14 TCP: этап SYN

- Начало трёхэтапного рукопожатия

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
94	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
95	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
96	1726.177517	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=
97	1727.182560	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 14503 → 7 [SYN] Seq=0
98	1727.183542	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
99	1727.198191	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=
100	1727.213884	192.168.1.12	192.168.1.11	ECHO	122	Request
101	1727.214868	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=57 Win=2920 Len=0
102	1727.244577	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920
103	1727.245579	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
104	1727.246576	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
105	1727.275400	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=

[Time shift for this packet: 0.00000000 seconds]
[Time delta from previous captured frame: 1.005043000 seconds]
[Time delta from previous displayed frame: 1.005043000 seconds]
[Time since reference or first frame: 28 minutes, 47.182560000 seconds]
Frame Number: 97

0000 00 50 79 66 68 00 00 50 79 66 68 01 00
0010 00 3c a5 29 00 00 40 06 52 2b c0 a8 00
0020 01 0b 38 a7 00 07 74 53 b8 37 00 00 00
0030 0b 68 48 99 00 00 02 04 05 b4 01 01 00
0040 a5 1a 00 00 00 00 01 03 03 01

14 TCP: этап SYN

- Начало трёхэтапного рукопожатия
- Флаг SYN

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
94	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
95	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
96	1726.177517	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=
97	1727.182560	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 14503 → 7 [SYN] Seq=0
98	1727.183542	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
99	1727.198191	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=
100	1727.213884	192.168.1.12	192.168.1.11	ECHO	122	Request
101	1727.214868	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=57 Win=2920 Len=0
102	1727.244577	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920
103	1727.245579	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
104	1727.246576	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
105	1727.275400	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=

[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.005043000 seconds]
[Time delta from previous displayed frame: 1.005043000 seconds]
[Time since reference or first frame: 28 minutes, 47.182560000 seconds]
Frame Number: 97

0000 00 50 79 66 68 00 00 50 79 66 68 01 00
0010 00 3c a5 29 00 00 40 06 52 2b c0 a8 00
0020 01 0b 38 a7 00 07 74 53 b8 37 00 00 00
0030 0b 68 48 99 00 00 02 04 05 b4 01 01 00
0040 a5 1a 00 00 00 00 01 03 03 01

14 TCP: этап SYN

- Начало трёхэтапного рукопожатия
- Флаг SYN
- Начальный Sequence Number

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
94	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
95	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
96	1726.177517	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=
97	1727.182560	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 14503 → 7 [SYN] Seq=0
98	1727.183542	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
99	1727.198191	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=
100	1727.213884	192.168.1.12	192.168.1.11	ECHO	122	Request
101	1727.214868	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=57 Win=2920 Len=0
102	1727.244577	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920
103	1727.245579	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
104	1727.246576	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
105	1727.275400	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=

[Time shift for this packet: 0.00000000 seconds]
[Time delta from previous captured frame: 1.005043000 seconds]
[Time delta from previous displayed frame: 1.005043000 seconds]
[Time since reference or first frame: 28 minutes, 47.182560000 seconds]
Frame Number: 97

0000 00 50 79 66 68 00 00 50 79 66 68 01 00
0010 00 3c a5 29 00 00 40 06 52 2b c0 a8 00
0020 01 0b 38 a7 00 07 74 53 b8 37 00 00 00
0030 0b 68 48 99 00 00 02 04 05 b4 01 01 00
0040 a5 1a 00 00 00 00 01 03 03 01

15 TCP: этап SYN-ACK

- Ответ принимающей стороны

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
94	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
95	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
96	1726.177517	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSva
97	1727.182560	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 14503 → 7 [SYN] Seq=0
98	1727.183542	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
99	1727.198191	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSva
100	1727.213884	192.168.1.12	192.168.1.11	ECHO	122	Request
101	1727.214868	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=57 Win=2920 Len=0
102	1727.244577	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920
103	1727.245579	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
104	1727.246576	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
105	1727.275400	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSva

▼ 000. = Flags: 0x0
0... = Reserved bit: Not set
.0.. = Don't fragment: Not set
..0. = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x522b [validation disabled]
[Header checksum status: Unverified]

0000 00 50 79 66 68 00 00 50 79 66 68 01
0010 00 3c a5 29 00 00 40 06 52 2b c0 a8
0020 01 0b 38 a7 00 07 74 53 b8 37 00 00
0030 0b 68 48 99 00 00 02 04 05 b4 01 01
0040 a5 1a 00 00 00 00 01 03 03 01

15 TCP: этап SYN-ACK

- Ответ принимающей стороны
- Подтверждение запроса

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
94	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
95	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
96	1726.177517	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSv
97	1727.182560	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 14503 → 7 [SYN] Seq=0
98	1727.183542	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
99	1727.198191	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSva
100	1727.213884	192.168.1.12	192.168.1.11	ECHO	122	Request
101	1727.214868	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=57 Win=2920 Len=0
102	1727.244577	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920
103	1727.245579	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
104	1727.246576	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
105	1727.275400	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSv

▼ 000. = Flags: 0x0
0... = Reserved bit: Not set
.0.. = Don't fragment: Not set
..0. = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x522b [validation disabled]
[Header checksum status: Unverified]

0000 00 50 79 66 68 00 00 50 79 66 68 01
0010 00 3c a5 29 00 00 40 06 52 2b c0 a8
0020 01 0b 38 a7 00 07 74 53 b8 37 00 00
0030 0b 68 48 99 00 00 02 04 05 b4 01 01
0040 a5 1a 00 00 00 00 01 03 03 01

16 TCP: этап АСК

- Завершение установки соединения

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
94	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
95	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
96	1726.177517	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSva
97	1727.182560	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 14503 → 7 [SYN] Seq=6
98	1727.183542	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
99	1727.198191	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSva
100	1727.213884	192.168.1.12	192.168.1.11	ECHO	122	Request
101	1727.214868	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=57 Win=2920 Len=0
102	1727.244577	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920
103	1727.245579	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
104	1727.246576	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
105	1727.275400	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSva

▼ 000. = Flags: 0x0
0... = Reserved bit: Not set
.0.. = Don't fragment: Not set
..0. = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x522b [validation disabled]
[Header checksum status: Unverified]

0000 00 50 79 66 68 00 00 50 79 66 68 01
0010 00 3c a5 29 00 00 40 06 52 2b c0 a8
0020 01 0b 38 a7 00 07 74 53 b8 37 00 00
0030 0b 68 48 99 00 00 02 04 05 b4 01 01
0040 a5 1a 00 00 00 00 01 03 03 01

16 TCP: этап АСК

- Завершение установки соединения
- TCP-соединение установлено

Capturing from Standard input [PC1-bahi Ethernet0 to msk-bahi-sw-01 Ethernet0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
94	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
95	1726.147961	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
96	1726.177517	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSva
97	1727.182560	192.168.1.12	192.168.1.11	TCP	74	[TCP Port numbers reused] 14503 → 7 [SYN] Seq=6
98	1727.183542	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0
99	1727.198191	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSva
100	1727.213884	192.168.1.12	192.168.1.11	ECHO	122	Request
101	1727.214868	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=57 Win=2920 Len=0
102	1727.244577	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920
103	1727.245579	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [ACK] Seq=1 Ack=58 Win=2920 Len=0
104	1727.246576	192.168.1.11	192.168.1.12	TCP	54	7 → 14503 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0
105	1727.275400	192.168.1.12	192.168.1.11	TCP	66	14503 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSva

▼ 000. = Flags: 0x0

- 0... .. = Reserved bit: Not set
- .0.. = Don't fragment: Not set
- ..0. = More fragments: Not set
- ...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: TCP (6)

Header Checksum: 0x522b [validation disabled]

[Header checksum status: Unverified]

0000 00 50 79 66 68 00 00 50 79 66 68 01

0010 00 3c a5 29 00 00 40 06 52 2b c0 a8

0020 01 0b 38 a7 00 07 74 53 b8 37 00 00

0030 0b 68 48 99 00 00 02 04 05 b4 01 01

0040 a5 1a 00 00 00 00 01 03 03 01

17 Топология с маршрутизатором FRR

- Добавлен маршрутизатор FRR



Рисунок 14: Топология сети с маршрутизатором FRR

17 Топология с маршрутизатором FRR

- Добавлен маршрутизатор FRR
- Включён захват трафика



Рисунок 14: Топология сети с маршрутизатором FRR

17 Топология с маршрутизатором FRR

- Добавлен маршрутизатор FRR
- Включён захват трафика
- Узлы переименованы



Рисунок 14: Топология сети с маршрутизатором FRR

18 IP-адресация PC-1 (FRR)

- Назначен адрес 192.168.1.10/24

```
PC1-bahi> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1-bahi : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1-bahi> save
Saving startup configuration to startup.vpc
: done

PC1-bahi> show ip

NAME          : PC1-bahi[1]
IP/MASK       : 192.168.1.10/24
GATEWAY       : 192.168.1.1
DNS           :
MAC           : 00:50:70:66:68:00
```

18 IP-адресация PC-1 (FRR)

- Назначен адрес 192.168.1.10/24
- Указан шлюз 192.168.1.1

```
PC1-bahi> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1-bahi : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

```
PC1-bahi> save
Saving startup configuration to startup.vpc
. done
```

```
PC1-bahi> show ip
```

```
NAME       : PC1-bahi[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:70:66:68:00
```

18 IP-адресация PC-1 (FRR)

- Назначен адрес 192.168.1.10/24
- Указан шлюз 192.168.1.1
- Проверка конфигурации

```
PC1-bahi> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1-bahi : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

```
PC1-bahi> save
Saving startup configuration to startup.vpc
: done
```

```
PC1-bahi> show ip
```

```
NAME       : PC1-bahi[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:70:66:68:00
```

19 IP-адресация FRR

- Настроен интерфейс маршрутизатора

```
vyos@msk-bahi-gw-01:~$ configure
[edit]
vyos@msk-bahi-gw-01# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# compare
[edit interfaces ethernet eth0]
+address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
Commit failed
[edit]
vyos@msk-bahi-gw-01# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@msk-bahi-gw-01# commit
[edit]
vyos@msk-bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-bahi-gw-01# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:9b:04:1e:00:00
  }
  ethernet eth1 {
    hw-id 0c:9b:04:1e:00:01
  }
  ethernet eth2 {
```

19 IP-адресация FRR

- Настроен интерфейс маршрутизатора
- Проверена конфигурация

```
vyos@msk-bahi-gw-01:~$ configure
[edit]
vyos@msk-bahi-gw-01# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# compare
[edit interfaces ethernet eth0]
+address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
Commit failed
[edit]
vyos@msk-bahi-gw-01# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@msk-bahi-gw-01# commit
[edit]
vyos@msk-bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-bahi-gw-01# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:9b:04:1e:00:00
  }
  ethernet eth1 {
    hw-id 0c:9b:04:1e:00:01
  }
  ethernet eth2 {
```

20 Проверка связи PC-1 ↔ FRR

- ICMP-эхо-запросы успешны

```
PC1-bahi> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.488 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=0.593 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=0.533 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=0.729 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.162 ms
```

```
PC1-bahi> █
```

20 Проверка связи PC-1 ↔ FRR

- ICMP-эхо-запросы успешны
- Связность подтверждена

```
PC1-bahi> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.488 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=0.593 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=0.533 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=0.729 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.162 ms
```

```
PC1-bahi> █
```

21 Анализ ICMP (FRR)

- Протокол: ICMP

Capturing from Standard input [msk-bahi-sw-01 Ethernet1 to msk-bahi-gw-01 eth0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

/Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::16	ICMPv6	138	Multicast Listener Report Message v2
2	0.202981	::	ff02::16	ICMPv6	138	Multicast Listener Report Message v2
3	0.299388	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x78f5b741
4	0.714896	::	ff02::1:ff1e:0	ICMPv6	86	Neighbor Solicitation for fe80::e9b:4ff:fe1e:0
5	1.740422	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
6	1.742824	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
7	1.907295	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
8	2.051006	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
9	3.011157	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x78f5b741
10	11.086091	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x78f5b741
11	22.875470	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.10 (Request)
12	23.876552	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.10 (Request)
13	24.414580	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x78f5b741
14	24.876793	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.10 (Request)
15	37.146379	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x78f5b741
16	50.113396	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x78f5b741
17	93.830022	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe920000e
18	122.748994	::	ff02::16	ICMPv6	138	Multicast Listener Report Message v2
19	123.018299	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
20	123.185985	::	ff02::16	ICMPv6	138	Multicast Listener Report Message v2
21	123.361674	::	ff02::1:ff1e:0	ICMPv6	86	Neighbor Solicitation for fe80::e9b:4ff:fe1e:0
22	124.387431	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
23	124.390755	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
24	124.961850	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
25	125.345779	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
26	126.700136	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
27	134.608193	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
28	148.956407	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
29	170.230075	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
30	260.516073	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x61522d71
31	265.375384	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x61522d71
32	276.129732	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x61522d71

21 Анализ ICMP (FRR)

- Протокол: ICMP
- MAC-адреса: unicast

Capturing from Standard input [msk-bahi-sw-01 Ethernet1 to msk-bahi-gw-01 eth0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

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5	1.740422	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
6	1.742824	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
7	1.907295	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
8	2.051006	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
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10	11.086091	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x78f5b741
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17	93.830022	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe920000e
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22	124.387431	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
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24	124.961850	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
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26	126.780136	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
27	134.608193	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
28	148.956407	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
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30	260.516073	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x61522d71
31	265.375384	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x61522d71
32	276.129732	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x61522d71

21 Анализ ICMP (FRR)

- Протокол: ICMP
- MAC-адреса: unicast
- IP-источник: PC-1

Capturing from Standard input [msk-bahi-sw-01 Ethernet1 to msk-bahi-gw-01 eth0]

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22	124.387431	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
23	124.390755	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
24	124.961850	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
25	125.345779	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
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21 Анализ ICMP (FRR)

- Протокол: ICMP
- MAC-адреса: unicast
- IP-источник: PC-1
- IP-назначение: FRR

Capturing from Standard input [msk-bahi-sw-01 Ethernet1 to msk-bahi-gw-01 eth0]

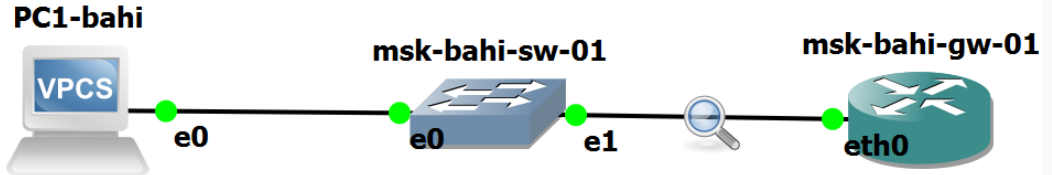
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7	1.907295	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
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20	123.185985	::	ff02::16	ICMPv6	138	Multicast Listener Report Message v2
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22	124.387431	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
23	124.390755	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
24	124.961850	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
25	125.345779	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
26	126.780136	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
27	134.608193	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5e951e1c
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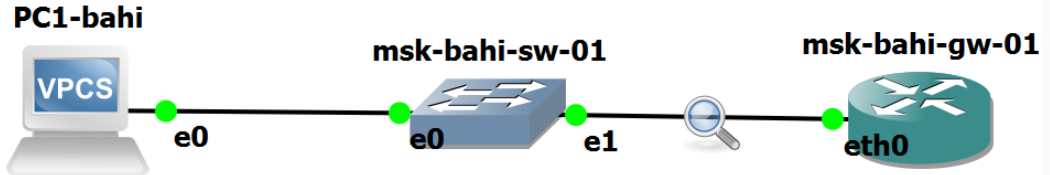
22 Топология с маршрутизатором VyOS

- Добавлен маршрутизатор VyOS



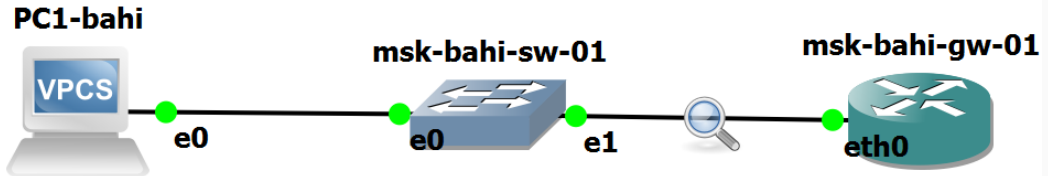
22 Топология с маршрутизатором VyOS

- Добавлен маршрутизатор VyOS
- Включён захват трафика



22 Топология с маршрутизатором VyOS

- Добавлен маршрутизатор VyOS
- Включён захват трафика
- Узлы запущены



23 IP-адресация PC-1 (VyOS)

- Назначен IPv4-адрес

```
PC1-bahi> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1-bahi : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

```
PC1-bahi> save
Saving startup configuration to startup.vpc
. done
```

```
PC1-bahi> show ip
```

```
NAME       : PC1-bahi[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500
```

23 IP-адресация PC-1 (VyOS)

- Назначен IPv4-адрес
- Подготовка к тестированию

```
PC1-bahi> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
PC1-bahi : 192.168.1.10 255.255.255.0 gateway 192.168.1.1
```

```
PC1-bahi> save
Saving startup configuration to startup.vpc
. done
```

```
PC1-bahi> show ip
```

```
NAME       : PC1-bahi[1]
IP/MASK     : 192.168.1.10/24
GATEWAY     : 192.168.1.1
DNS         :
MAC         : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT  : 127.0.0.1:20005
MTU         : 1500
```

24 Вход в систему VyOS

- Login: vyos

```
[ 13.459365] mpls_gso: MPLS GSO support
```

```
Welcome to VyOS - msk-bahi-gw-01 ttyS0
```

```
msk-bahi-gw-01 login: vyos
```

```
Password:
```

```
Welcome to VyOS!
```

```
Check out project news at https://blog.vyos.io  
and feel free to report bugs at https://vyos.dev
```

```
You can change this banner using "set system login banner post-login" command.
```

```
VyOS is a free software distribution that includes multiple components,  
you can check individual component licenses under /usr/share/doc/*/copyright  
vyos@msk-bahi-gw-01:~$ █
```

24 Вход в систему VyOS

- >Login: vyos
- Пароль: vyos

```
[ 13.459365] mpls_gso: MPLS GSO support
```

```
Welcome to VyOS - msk-bahi-gw-01 ttyS0
```

```
msk-bahi-gw-01 login: vyos
```

```
Password:
```

```
Welcome to VyOS!
```

```
Check out project news at https://blog.vyos.io  
and feel free to report bugs at https://vyos.dev
```

```
You can change this banner using "set system login banner post-login" command.
```

```
VyOS is a free software distribution that includes multiple components,  
you can check individual component licenses under /usr/share/doc/*/copyright  
vyos@msk-bahi-gw-01:~$ █
```

24 Вход в систему VyOS

- >Login: vyos
- Пароль: vyos
- Переход в CLI

```
[ 13.459365] mpls_gso: MPLS GSO support
```

```
Welcome to VyOS - msk-bahi-gw-01 ttyS0
```

```
msk-bahi-gw-01 login: vyos
```

```
Password:
```

```
Welcome to VyOS!
```

```
Check out project news at https://blog.vyos.io  
and feel free to report bugs at https://vyos.dev
```

```
You can change this banner using "set system login banner post-login" command.
```

```
VyOS is a free software distribution that includes multiple components,  
you can check individual component licenses under /usr/share/doc/*/copyright  
vyos@msk-bahi-gw-01:~$ █
```

25 Настройка интерфейса eth0 (VyOS)

- Удалён DHCP

```
vyos@msk-bahi-gw-01:~$ configure
[edit]
vyos@msk-bahi-gw-01# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# compare
[edit interfaces ethernet eth0]
+address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
Commit failed
[edit]
vyos@msk-bahi-gw-01# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@msk-bahi-gw-01# commit
[edit]
vyos@msk-bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-bahi-gw-01# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:9b:04:1e:00:00
  }
  ethernet eth1 {
    hw-id 0c:9b:04:1e:00:01
```

25 Настройка интерфейса eth0 (VyOS)

- Удалён DHCP
- Назначен адрес 192.168.1.1/24

```
vyos@msk-bahi-gw-01:~$ configure
[edit]
vyos@msk-bahi-gw-01# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# compare
[edit interfaces ethernet eth0]
+address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
Commit failed
[edit]
vyos@msk-bahi-gw-01# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@msk-bahi-gw-01# commit
[edit]
vyos@msk-bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-bahi-gw-01# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:9b:04:1e:00:00
  }
  ethernet eth1 {
    hw-id 0c:9b:04:1e:00:01
```

25 Настройка интерфейса eth0 (VyOS)

- Удалён DHCP
- Назначен адрес 192.168.1.1/24
- Выполнены `commit` и `save`

```
vyos@msk-bahi-gw-01:~$ configure
[edit]
vyos@msk-bahi-gw-01# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# compare
[edit interfaces ethernet eth0]
+address 192.168.1.1/24
[edit]
vyos@msk-bahi-gw-01# commit
```

Can't configure both static IPv4 and DHCP address on the same interface

```
[[interfaces ethernet eth0]] failed
Commit failed
[edit]
vyos@msk-bahi-gw-01# delete interfaces ethernet eth0 address dhcp
[edit]
vyos@msk-bahi-gw-01# commit
[edit]
vyos@msk-bahi-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-bahi-gw-01# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:9b:04:1e:00:00
  }
  ethernet eth1 {
    hw-id 0c:9b:04:1e:00:01
```

26 Проверка связи PC-1 ↔ VyOS

- Выполнен ping

```
PC1-bahi> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.488 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=0.593 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=0.533 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=0.729 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.162 ms
```

```
PC1-bahi> █
```

26 Проверка связи PC-1 ↔ VyOS

- Выполнен ping
- Получены 5 ICMP-ответов

```
PC1-bahi> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.488 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=0.593 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=0.533 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=0.729 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.162 ms
```

```
PC1-bahi> █
```

27 Анализ ICMP (VyOS)

- Протокол: ICMP

Capturing from Standard input [msk-bahi-sw-01 Ethernet1 to msk-bahi-gw-01 eth0]

No.	Time	Source	Destination	Protocol	Length	Info
100	1543.207452	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
101	1547.643580	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
102	1557.103449	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
103	1577.640374	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
104	1588.522055	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
105	1600.499586	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
106	1675.768471	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
107	1681.205609	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
108	1688.542370	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
109	1696.919340	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
110	1715.401772	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
111	1731.803881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
112	1784.733156	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
113	1786.103051	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
114	1788.612729	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
115	1789.988144	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
116	1794.147663	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
117	1798.127120	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
118	1807.920793	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
119	1839.787131	::	ff02::16	ICMPv6	130	Multicast Listener Report Message v2
120	1839.936136	::	ff02::1:ff1e:0	ICMPv6	86	Neighbor Solicitation for fe80::e9b:4ff:fe1e:0
121	1840.025047	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
122	1840.231317	::	ff02::16	ICMPv6	130	Multicast Listener Report Message v2
123	1840.960523	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
124	1840.963317	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
125	1841.663351	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
126	1841.727400	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
127	1843.062057	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
128	1850.443070	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
129	1871.441506	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
130	1880.096899	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
131	1890.460835	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
132	1898.609421	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
133	1935.645784	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x10dda400
134	1938.030671	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x10dda400

27 Анализ ICMP (VyOS)

- Протокол: ICMP
- Источник: PC-1

Capturing from Standard input [msk-bahi-sw-01 Ethernet1 to msk-bahi-gw-01 eth0]

No.	Time	Source	Destination	Protocol	Length	Info
100	1543.207452	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
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102	1557.103449	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
103	1577.640374	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
104	1588.522055	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
105	1600.499586	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
106	1675.768471	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
107	1681.205609	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
108	1688.542370	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
109	1696.919340	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
110	1715.401772	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
111	1731.803881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
112	1784.733156	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
113	1786.103051	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
114	1788.612729	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
115	1789.988144	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
116	1794.147663	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
117	1798.127120	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
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120	1839.936136	::	ff02::1:ff1e:0	ICMPv6	86	Neighbor Solicitation for fe80::e9b:4ff:fe1e:0
121	1840.025047	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
122	1840.231317	::	ff02::16	ICMPv6	130	Multicast Listener Report Message v2
123	1840.960523	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
124	1840.963317	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
125	1841.663351	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
126	1841.727400	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
127	1843.062057	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
128	1850.443070	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
129	1871.441506	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
130	1880.096899	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
131	1890.460835	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
132	1898.609421	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
133	1935.645784	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x10dda400
134	1938.030671	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x10dda400

27 Анализ ICMP (VyOS)

- Протокол: ICMP
- Источник: PC-1
- Назначение: VyOS

Capturing from Standard input [msk-bahi-sw-01 Ethernet1 to msk-bahi-gw-01 eth0]

No.	Time	Source	Destination	Protocol	Length	Info
100	1543.207452	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
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103	1577.640374	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
104	1588.522055	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
105	1600.499586	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xb120a27d
106	1675.768471	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
107	1681.205609	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
108	1688.542370	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
109	1696.919340	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
110	1715.401772	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
111	1731.803881	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x6946da48
112	1784.733156	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
113	1786.103051	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
114	1788.612729	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
115	1789.988144	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
116	1794.147663	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
117	1798.127120	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
118	1807.920793	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x8161cf43
119	1839.787131	::	ff02::16	ICMPv6	130	Multicast Listener Report Message v2
120	1839.936136	::	ff02::1:ff1e:0	ICMPv6	86	Neighbor Solicitation for fe80::e9b:4ff:fe1e:0
121	1840.025047	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
122	1840.231317	::	ff02::16	ICMPv6	130	Multicast Listener Report Message v2
123	1840.960523	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
124	1840.963317	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
125	1841.663351	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
126	1841.727400	fe80::e9b:4ff:fe1e:0	ff02::16	ICMPv6	150	Multicast Listener Report Message v2
127	1843.062057	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
128	1850.443070	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
129	1871.441506	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
130	1880.096899	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
131	1890.460835	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
132	1898.609421	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7004611b
133	1935.645784	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x10dda400
134	1938.030671	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x10dda400

- Смоделированы сети с FRR и VyOS

- Смоделированы сети с FRR и VyOS
- Проверена IP-связность

- Смоделированы сети с FRR и VyOS
- Проверена IP-связность
- Проанализированы ARP, ICMP, UDP, TCP

- Смоделированы сети с FRR и VyOS
- Проверена IP-связность
- Проанализированы ARP, ICMP, UDP, TCP
- Подтверждена корректность конфигурации